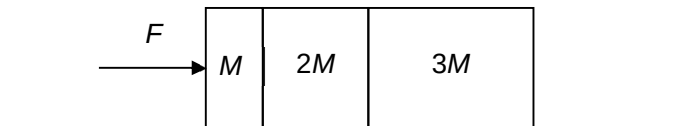


- 6 Three blocks with masses M , $2M$ and $3M$ are pushed along a rough surface by a horizontal force F as shown.



The frictional force between a block and the rough surface is directly proportional to the weight of the block. Given that the frictional force between the block with mass M and the rough surface is f , what is the magnitude of the force mass $3M$ exerts on mass $2M$?

- A** $\frac{F}{2}$
- B** $\frac{F}{2} + 3f$
- C** $\frac{F}{3} + f$
- D** $\frac{F}{3} + 3f$

Consider the whole system,

$$F - 6f = 6Ma$$

$$a = \frac{F - 6f}{6M}$$

Let the force that $3M$ acts on $2M$ be F_1 .

Consider Newton's 2nd law on $3M$,

$$F_1 - 3f = 3Ma$$

$$F_1 = 3M \left(\frac{F - 6f}{6M} \right) + 3f = \frac{F}{2}$$

- 7 A uniform rod of length $4a$ is supported by a hinge at one end and a force of 100 N at Y and Z . If the force